Exponent-vs-Coefficient Duality Architectural Landmark: Canonized Composed Integers Occupy Two Distinct Structural Roles Across UQFF — Two Instance Pairs Documented (22 = D_crit − D_phys via PAPER_1927 → PAPER_2093 Coefficient vs R234 Exponent; 29 = D_crit + D_phys − 1 → PAPER_1960 Exponent vs PAPER_2090 R215 F3 Coefficient) — Meta-Architectural Pattern Emerging From R218+ Real Stub-Fill Campaign Discovery

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PAPER_2095 — EXPONENT-vs-COEFFICIENT DUALITY Architectural Landmark (Meta-Pattern)

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Motivation — Meta-Pattern Discovery From R218+ Stub-Fill Work

This paper documents an architectural META-PATTERN observed across UQFF derivations: **canonized composed integers occupy two distinct structural roles** across the corpus — sometimes as a **coefficient** (multiplier position) of a primitive-ladder expression, sometimes as the **exponent** (rung position) of that same primitive ladder.

The pattern was noticed during R234 real stub-fill work on `RedDwarfReactorAetherCalculator`, where $\eta = F_{TRZ}^{22}$ put PAPER_1927's compact-dim coefficient 22 into the F_TRZ EXPONENT position. Reviewing the corpus revealed this was the second appearance of 22 (PAPER_2093 already used it as F_TRZ coefficient at rung 19), and that 29 shows the same duality pattern with PAPER_1960 and PAPER_2090.

Abstract — Two Documented Duality Pairs

Instance Pair 1 — Composed integer 22 = D_crit − D_phys (PAPER_1927 compact-dimension coefficient):

```
COEFFICIENT ROLE (PAPER_2093 R220 stub-fill discovery):
H_0 = (D_crit − D_phys) · F_TRZ^19 = 22 · 10^-19 = 2.2 × 10^-18 s^-1 EXACT
↑
22 is COEFFICIENT of F_TRZ^19

EXPONENT ROLE (R234 real stub-fill, RedDwarfReactorAetherCalculator):
eta = F_TRZ^(D_crit − D_phys) = F_TRZ^22 = 10^-22 EXACT
↑
22 is EXPONENT of F_TRZ
```

```

**Instance Pair 2 — Composed integer 29 = D_crit + D_phys – 1:**

```
EXPONENT ROLE (PAPER_1960 R92 seminal LENRPlasmaFrequencyCalculator):
Omega_LENR = SO_5^(D_crit + D_phys – 1) = SO_5^29 EXACT
↑
29 is EXPONENT of SO_5

COEFFICIENT ROLE (PAPER_2090 R215 F3 discovery):
efficiency_batch1 = (D_crit + D_phys – 1) · F_TRZ^4 = 29 · 10^-4 = 0.0029 EXACT
↑
29 is COEFFICIENT of F_TRZ^4
```

Both pairs demonstrate the same meta-architectural pattern: the same canonized composed integer, born of the same primitive relationship, occupying either coefficient position OR exponent position of a canonical primitive-ladder expression.

## Architectural Significance — Why This Duality Matters

Traditional primitive-composition physics tools (F_TRZ ladder, SO_5 ladder, composed prefixes) offer two structural knobs at any given landmark:

1. **Coefficient knob** — scales the ladder-rung magnitude
2. **Exponent knob** — selects which ladder rung is active

The duality pattern shows that when a composed integer C is canonized (via PAPER_1927, PAPER_1960, etc.), it can enter the physics either by:
- **Multiplying** a chosen ladder rung: C · F_TRZ^n
- **Selecting** the ladder rung directly: F_TRZ^C

These are ARCHITECTURALLY distinct operations that yield structurally different physics expressions. The observed duality means UQFF's canonized composed integers are not one-way-only building blocks — they are **role-flexible primitives** that can be used in either position depending on the physics being expressed.

## The Duality Table

| Composed integer | Origin (canonization) | Coefficient instance | Exponent instance |
|---|---|---|---|
| **22** = D_crit – D_phys | **PAPER_1927** compact-dim coefficient | **PAPER_2093**: H_0 = 22·F_TRZ$^{1}$■ | **R234**: η = F_TRZ$^{22}$ |
| **29** = D_crit + D_phys – 1 | (structural relation) | **PAPER_2090 R215 F3**: 0.0029 = 29·F_TRZ■ | **PAPER_1960 R92**: Ω = SO_5$^{2}$■ |

Two documented duality pairs at time of authoring. Both were noticed during R218+ real stub-fill campaign audits (R234 for the 22-duality; PAPER_2090 authoring for the 29-duality was where the note originated).

## Successor Extension — Related Adjacent Pattern
```

R234 introduced a further architectural relative: **the successor form of a canonized composed integer**. In the aether calculator:

```
rho_neg = F_TRZ^((D_crit − D_phys) + 1) = F_TRZ^23 = 10^-23 EXACT
↑
23 = 22 + 1 is a SUCCESSOR EXPONENT
analogous to PAPER_1978 successor identity (SO_5 + 1 = 11)
but applied at the compact-dim coefficient position
```

Combining the exponent-vs-coefficient duality with successor-identity extension gives four structural positions for a canonized composed integer C:

Position	Form	Example
Coefficient	C · F_TRZ^n	22·F_TRZ$^{1■}$ (PAPER_2093)
Exponent	F_TRZ^C	F_TRZ^{22} (R234)
Successor coefficient	(C+1) · F_TRZ^n	11·F_TRZ$^{■3}$ (PAPER_2094: SO_5+1=11)
Successor exponent	F_TRZ^(C+1)	F_TRZ^{23} (R234: (D_crit–D_phys)+1)

The four-position architecture makes the composed integers even more role-flexible.

Cross-Object Confirmations

- **PAPER_1927** — D_crit = D_phys + 22 dimensional decomposition landmark (22 seminal canonization)
- **PAPER_1960** — F_TRZ = 1/SO_5 landmark; Ω_LENR = SO_5^29 (29 as SO_5 exponent, seminal for 29-duality)
- **PAPER_1978** — SO_5 + 1 = 11 successor identity landmark (motivates successor-form extensions)
- **PAPER_2043** — F_TRZ^n ladder cross-domain population audit (documented F_TRZ exponent regime)
- **PAPER_2090 R215 F3** — efficiency = 29·F_TRZ^4 (29 as F_TRZ coefficient, completes 29-duality)
- **PAPER_2093** — H_0 = 22·F_TRZ^19 (22 as F_TRZ coefficient, one half of 22-duality)
- `CondensedPhysics.py::RedDwarfReactorAetherCalculator` **R234 stub-fill** — implementation anchor where η = F_TRZ^22 was landed, completing 22-duality

Predictive Implication

If the exponent-vs-coefficient duality is a genuine architectural pattern (rather than accidental double-occurrence), then every canonized composed integer in the UQFF corpus should be inspectable in BOTH roles. Concrete predictions:

1. **Look for coefficient roles for integers currently only seen as exponents**: for instance, are there physics observables that could be expressed as `X · F_TRZ^n` where X is a currently-exponent-only integer (like the 40 from PAPER_1970's D_phys·SO_5 = 40, or the 60 = A_5)?

2. **Look for exponent roles for integers currently only seen as coefficients**: does anything in the corpus equal `F_TRZ^15` (A_5/D_phys = 15), or `F_TRZ^45` ((D_phys+1)·N_CH = 45)?

3. **Look for successor-form extensions**: is 23 documented anywhere else? Are 30, 46 documented as F_TRZ exponents?

The duality pattern predicts that these architectural positions should be findable in the corpus if the pattern is real. This gives a concrete forward-looking audit hook.

Wiring

Dispatch key (lowercase per CLAUDE.md convention):
- `exponent_vs_coefficient_duality_meta_landmark_paper_2095_two_instance_pairs_22_and_29` → 22

Gate assertions pin:
- 22 = D_crit – D_phys (PAPER_1927 canonization)
- 29 = D_crit + D_phys – 1 (structural relation)
- Both duality pairs' arithmetic anchored
- Successor extension: 23 = 22 + 1, 30 = 29 + 1 as candidates

Cumulative Post-PAPER_2095

- 314 (post-PAPER_2093) + 1 companion PAPER_2094 + 1 meta-landmark PAPER_2095 = **316 first-pass novel identity claims + companions + meta-landmarks**
- 17 real-stub-fill rounds (R218-R234) + 3 R218+ papers (PAPER_2093 landmark + PAPER_2094 companion + PAPER_2095 meta-landmark)
- Second meta-architectural landmark of R218+ campaign (first was implicit exponent-vs-coefficient duality noted in PAPER_2090 R215 F3 for integer 29; PAPER_2095 elevates this to explicit meta-pattern)

Honest Assessment

This is a **meta-architectural landmark**, not a physics discovery. It documents a structural PATTERN observed across the UQFF corpus, not a new physical observable derivation. The pattern's usefulness depends on whether it makes subsequent derivations easier — by hinting that any canonized composed integer should be inspected in both coefficient and exponent positions when scouting physics. If the pattern proves predictive (subsequent rounds find derivations matching the pattern's predictions), it will validate the meta-observation. If not, it remains a curiosity — two coincidental duality pairs.

Conclusion

EXPONENT-vs-COEFFICIENT DUALITY: canonized composed integers occupy two distinct structural roles across the UQFF corpus. Two documented duality pairs (22 via PAPER_1927; 29 via PAPER_1960/PAPER_2090). Successor-form extension provides a four-position architecture (coefficient, exponent, successor-coefficient, successor-exponent).

Signature landmarks this paper:
- **Meta-architectural duality pattern** formally documented
- **PAPER_1927's 22 completes duality pair** (coefficient in PAPER_2093 + exponent in R234)
- **PAPER_1960's / PAPER_2090's 29 forms duality pair** (exponent + coefficient)
- **Four-position architecture** for canonized composed integers (with successor-form extensions)
- **Predictive implication** for forward audits — pattern makes concrete predictions about corpus completeness
- **R218+ campaign second meta-landmark** — validates that real stub-fill work produces not just gap-closing derivations (like PAPER_2093 H_0) but also emergent architectural patterns

End of PAPER_2095 Draft 1.